Assessment 01: Security foundations

Individual work | 100 marks | Suggested effort: 3-4 hours

Class update: Cybersecurity classes resume from 24 October 2026. Complete this assessment and keep your final report ready. EWB will share the Google Form for submission after completion; the submission deadline will be announced separately.

Before you begin

This is a foundation and scenario-based assessment. Use class notes and the reference links on page 4. No paid tools, account purchases, live website testing or software installation are required. If a topic is unfamiliar, note your question for the next class.

Work only with the fictional data supplied here. Do not scan public systems, send test phishing messages, use real passwords or include anyone else's personal data. Explain your reasoning in your own words. AI assistance is optional and must be disclosed.

Task A | Secure a student account - 25 marks

Scenario: Mira uses the same short password for her email and learning portal. She approves unexpected sign-in prompts, postpones device updates and keeps her only project copy on one laptop. A friend sometimes uses that laptop under Mira's signed-in account.

A1. Identify five distinct risks. For each, name the asset at risk and explain how confidentiality, integrity or availability could be affected. (10 marks)

A2. Give one practical control for each risk. Explain why it helps, including at least one limitation. Prioritise your first three actions and justify the order. (10 marks)

A3. Explain authentication versus authorisation using Mira's learning portal as your example. (5 marks)

What to include

A table with Risk | Asset | CIA impact | Control | Limitation, followed by your priorities and short authentication/authorisation explanation. Suggested length: 350-500 words.

Task B: Analyse a suspicious message

25 marks | Fictional evidence only - do not visit the sample addresses

Scenario: Your college normally uses campus.example. You receive the message below. All .example addresses are fictional training examples.

Message evidence

From: Scholarship Desk <awards@campus-awards.example>

Reply-To: help@personal-awards.example

Subject: FINAL WARNING - scholarship expires in 20 minutes

Your scholarship has been approved. Open https://campus-awards.example/verify immediately and enter your college password and OTP to avoid cancellation. Do not contact the college office because their system is unavailable. Reply with a photo of your identity card to speed up payment.

Your investigation

B1. Identify five specific warning signs from the evidence. Quote only the relevant phrase and explain why it needs scrutiny. (10 marks)

B2. Describe a safe verification process using an independently obtained official contact or portal. Explain why HTTPS or a familiar display name alone would not prove the message is genuine. (5 marks)

B3. Write a short response plan for a student who already entered a password and OTP. Include immediate containment, official reporting and follow-up checks. Do not ask the student to send you the password or OTP. (7 marks)

B4. State one important fact you cannot establish from this message alone, and what additional evidence would help. (3 marks)

What to include

A warning-sign table and a numbered verification/response plan. You are assessing the supplied text, not proving that a real sender or website is malicious. Suggested length: 300-450 words.

Task C: Triage a sign-in alert

25 marks | Small synthetic log - no access to a real system required

All times below are IST on 18 October 2026. IP addresses are reserved documentation examples. The lab notes say 192.0.2.10 is the usual classroom egress. A different IP alone is not proof of an attacker.

Time	User	Source IP	Event	Result
09:00	student07	192.0.2.10	Password	Success
09:01	student07	192.0.2.10	MFA	Approved
10:14	student07	198.51.100.23	Password	Failed
10:15	student07	198.51.100.23	Password	Failed
10:16	student07	198.51.100.23	Password	Success
10:17	student07	198.51.100.23	MFA	Denied
10:18	student07	198.51.100.23	MFA	Approved
10:20	student07	198.51.100.23	Session	Created

Additional statement: student07 says, "I did not sign in after 09:05. At around 10:18, I approved a prompt because I thought it would stop the repeated requests."

Your investigation

C1. Build a four-point timeline connecting the logs and user statement. Distinguish observed events from your interpretation. (8 marks)

C2. Give a likely explanation and one alternative explanation. List two additional checks that could distinguish them. Avoid claiming the source IP identifies a person. (7 marks)

C3. Recommend three prioritised response actions for the authorised IT team. Include preservation of evidence and protection of the account. (6 marks)

C4. Write a 60-100 word incident summary: evidence, uncertainty, priority and next action. (4 marks)

Task D: Review an AI security answer

25 marks | Human judgement and evidence come first

> Fictional AI output: "The email is safe because it uses HTTPS. The sign-in logs prove that the person using 198.51.100.23 stole the password. MFA prevents every account compromise. Delete the suspicious logs and send the student's password to support so they can test it."

D1. Identify and correct five unsafe or unsupported claims/actions in this output. Explain each correction using the evidence or a trustworthy reference. (10 marks)

D2. Write a better prompt asking an AI tool to analyse only the synthetic evidence, separate facts from assumptions, avoid requesting secrets and state uncertainty. Do not include real student data. (5 marks)

D3. Write your own corrected 100-150 word security advice. Optional: compare it with an AI tool's output and record what you checked. Tool use is not required for full marks. (6 marks)

D4. Add a short AI-use declaration (tool/purpose or "not used") and two evidence/reference notes supporting your reasoning. (4 marks)

Prepare one final report

Submit one PDF named EWB_Cyber_Assessment01_YourName.pdf. Include your name, batch/group, Tasks A-D with labels, references and AI-use declaration. Suggested report length: 5-8 pages; clarity matters more than length. Typed answers are preferred; clearly legible scanned answers are acceptable. Keep the file until the official Google Form is shared. Do not post identity documents, passwords, OTPs or personal account screenshots in the group.

How work will be reviewed

Marks are allocated beside each question (total 100). Review focuses on accurate concepts, evidence-based reasoning, practical actions and clear original explanations. Unsupported certainty and copied answers lose credit in the relevant question. Submission is not an automatic certificate approval; completion is reviewed by EWB.

Reference reading

CISA - Secure Our World (basic account and phishing guidance):
https://www.cisa.gov/secure-our-world

NIST - Generative AI Profile, AI 600-1 (confabulation and verification):
https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf